Distributed Optimization with Anytime Coupling Constraints and Asymptotic Consensus

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***Abstract*—In this paper, we study distributed consensus optimization with anytime coupling constraints, meaning that the coupling inequalities remain satisfied throughout the algorithm trajectory, while consensus is required only asymptotically. A continuous-time distributed method is developed by combining projected saddle-point dynamics with local safe gradient-flow updates. We show that, under the stated regularity conditions, the algorithm can converge arbitrarily close to the optimizer, while exact convergence is guaranteed under additional boundedness conditions. A numerical example illustrates the theoretical results.**

***Index Terms*—Distributed optimization, anytime feasibility, asymptotic consensus, coupling constraints, safe gradient flow.**

I. Introduction

Distributed optimization provides a basic computational framework for networked control, estimation, resource allocation, and multi-agent coordination [1], [2]. Each agent uses only local objective or constraint information and communicates with its neighbors. Continuous-time algorithms are closely connected with coordination dynamics and can be integrated with control systems [3], [4]. Constrained variants include primal-dual and projected methods for local, set, equality, and coupled constraints [5], [6]. Most of these methods reach a feasible optimizer asymptotically, but their transient trajectories may violate constraints with immediate operational meaning.

This issue is important when the algorithmic state is used online as a resource allocation, control input, or reference signal. Recent feasible or violation-free distributed methods preserve the relevant constraints along the entire trajectory [7], [8]. Related ideas arise in safety-critical control, where barrier functions enforce forward invariance of safe sets [9]. Safe gradient flow (SGF) constructs such a vector field through a quadratic program and links its equilibria to Karush–Kuhn–Tucker (KKT) points [10]. Building on projected saddle-point convergence results [11], Mestres and Cortés [12] proposed a distributed anytime algorithm for separable network optimization.

In consensus optimization, coupling inequalities and consensus equations play different roles. The former may represent operating limits that must hold throughout the algorithm, whereas consensus need only be achieved asymptotically. Enforcing consensus as an anytime equality forces the output to start from a consensual state. Motivated by this distinction, we combine a regularized projected saddle-point subsystem with local SGFs for the localized inequalities, while Laplacian primal and multiplier couplings drive the disagreement to zero.

The main contributions are summarized as follows.

(i) We extend the anytime framework of [12] by separating inequality preservation from consensus enforcement. The local SGF subproblems impose only localized inequalities, while the nominal Laplacian dynamics achieve consensus. Therefore, the output need not be initialized consensually.

(ii) The rank-deficient Laplacian makes the equality multiplier nonunique, preventing direct use of existing cascade results. We remove this ambiguity in reduced coordinates, prove auxiliary and cascade convergence, and bound the regularization error.

Notations. The sets of real and nonnegative real numbers are denoted by $\mathbb{R}$ and $\mathbb{R}_{\geq 0}$, respectively. The symbol $\|\cdot\|$ denotes the Euclidean norm. Moreover, $\mathrm{col}(\boldsymbol{z}_1, \ldots, \boldsymbol{z}_N)$ denotes the column concatenation of its arguments, $\mathbf{1}_N$ the all-ones vector in $\mathbb{R}^N$, I_n the $n \times n$ identity matrix, and $\otimes$ the Kronecker product. For a matrix A, $\mathrm{Range}(A)$ and $\ker(A)$ denote its range and null space. For a subspace $\mathcal{S}$, $\mathcal{S}^\perp$ is its orthogonal complement, and for a symmetric matrix M, $M \succeq 0$ means that M is positive semidefinite. For $r \in \mathbb{R}$ and $\lambda \in \mathbb{R}_{\geq 0}$, $[r]_\lambda^+ := r$ when $\lambda > 0$ and $[r]_\lambda^+ := \max\{0, r\}$ when $\lambda = 0$. For an undirected weighted graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, $\mathcal{N}_i := \{j \in \mathcal{V} : (i,j) \in \mathcal{E}\}$ is the neighbor set of node i. The weights satisfy $a_{ij} = a_{ji} > 0$ for $(i,j) \in \mathcal{E}$ and $a_{ij} = 0$ otherwise. Its weighted Laplacian is $L \in \mathbb{R}^{N \times N}$, with $L_{ii} := \sum_{j \in \mathcal{N}_i} a_{ij}$ and $L_{ij} := -a_{ij}$ for $i \neq j$, and $\mathcal{L} := L \otimes I_n$ is the lifted Laplacian.

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For $\boldsymbol{z} = \mathrm{col}(\boldsymbol{z}_1, \dots, \boldsymbol{z}_N)$, $\boldsymbol{z}_{\mathcal{N}_i}$ collects $\boldsymbol{z}_j$ over $j \in \mathcal{N}_i$.

II. Problem Formulation and Preliminaries

Consider a network of N agents over $\mathcal{G}$. Agent i holds $\boldsymbol{x}_i \in \mathbb{R}^n$, and $\boldsymbol{x} := \mathrm{col}(\boldsymbol{x}_1, \dots, \boldsymbol{x}_N) \in \mathbb{R}^{Nn}$. Define $F(\boldsymbol{x}) := \sum_{i=1}^N f_i(\boldsymbol{x}_i)$ and $G_k(\boldsymbol{x}) := \sum_{i=1}^N g_i^k(\boldsymbol{x}_i)$, $k = 1, \dots, p$. We consider

$$\begin{aligned} \min_{\boldsymbol{x}_1, \dots, \boldsymbol{x}_N} \quad & F(\boldsymbol{x}) \\ \text{s.t.} \quad & G_k(\boldsymbol{x}) \le 0, \quad k = 1, \dots, p, \\ & \boldsymbol{x}_1 = \cdots = \boldsymbol{x}_N. \end{aligned} \tag{P}$$

When $\mathcal{G}$ is connected, the consensus constraint in (P) is equivalent to $\mathcal{L}\boldsymbol{x} = \boldsymbol{0}$. The coupling-feasible set is

$$\mathcal{X}_G := \{\boldsymbol{x} \in \mathbb{R}^{Nn} : G_k(\boldsymbol{x}) \le 0,\ k = 1, \dots, p\}.$$

We require anytime inequalities, i.e., $G_k(\boldsymbol{x}(0)) \le 0$ implies $G_k(\boldsymbol{x}(t)) \le 0$ for every $t \ge 0$, and asymptotic consensus, $\mathcal{L}\boldsymbol{x}(t) \to \boldsymbol{0}$. Unlike an anytime consensus equality, the latter does not require $\mathcal{L}\boldsymbol{x}(0) = \boldsymbol{0}$.

We impose the following standard assumptions.

Assumption 1 (Graph)**.** *The communication graph $\mathcal{G}$ is undirected and connected.*

Under Assumption 1, $\boldsymbol{1}_N^\top L = \boldsymbol{0}^\top$, $\ker(L) = \mathrm{span}\{\boldsymbol{1}_N\}$, and $\mathrm{Range}(L) = \boldsymbol{1}_N^\perp$.

Assumption 2 (Convexity and smoothness)**.** *For each $i \in \mathcal{V}$, f_i is continuously differentiable, σ-strongly convex for some $\sigma > 0$, and has locally Lipschitz gradient. For each $i \in \mathcal{V}$ and $k = 1, \dots, p$, g_i^k is continuously differentiable, convex, and has locally Lipschitz gradient.*

Assumption 3 (Strict feasibility)**.** *There exists $\bar{\boldsymbol{z}} \in \mathbb{R}^n$ such that $\sum_{i=1}^N g_i^k(\bar{\boldsymbol{z}}) < 0$ for $k = 1, \dots, p$.*

For later use, let $\phi, q_\ell : \mathbb{R}^n \to \mathbb{R}$, $\ell = 1, \dots, m$, and consider the inequality-constrained problem

$$\begin{aligned} \min_{\boldsymbol{z} \in \mathbb{R}^n} \quad & \phi(\boldsymbol{z}) \\ \text{s.t.} \quad & q_\ell(\boldsymbol{z}) \le 0, \quad \ell = 1, \dots, m. \end{aligned} \tag{1}$$

Assume these functions have locally Lipschitz gradients. Let $\mathcal{C} := \{\boldsymbol{z} \in \mathbb{R}^n : q_\ell(\boldsymbol{z}) \le 0,\ \ell = 1, \dots, m\}$ and $\mathcal{I}(\boldsymbol{z}) := \{\ell : q_\ell(\boldsymbol{z}) = 0\}$. LICQ means that $\{\nabla q_\ell(\boldsymbol{z}) : \ell \in \mathcal{I}(\boldsymbol{z})\}$ is linearly independent. MFCQ means that some $\boldsymbol{d}$ satisfies $\nabla q_\ell(\boldsymbol{z})^\top \boldsymbol{d} < 0$ for every active ℓ. For $\alpha > 0$, the SGF associated with (1) is

$$\begin{aligned} \mathcal{S}_\alpha(\boldsymbol{z}) := \arg\min_{\boldsymbol{\xi} \in \mathbb{R}^n} \quad & \frac{1}{2}\|\boldsymbol{\xi} + \nabla\phi(\boldsymbol{z})\|^2 \\ \text{s.t.} \quad & \nabla q_\ell(\boldsymbol{z})^\top \boldsymbol{\xi} \le -\alpha q_\ell(\boldsymbol{z}), \quad \ell = 1, \dots, m. \end{aligned} \tag{2}$$

We refer to the quadratic program in (2) as SGF-QP.

Proposition 1. *Suppose MFCQ holds on $\mathcal{C}$ and $\mathcal{S}_\alpha$ is locally Lipschitz on a neighborhood of $\mathcal{C}$. Then the following hold:*

(i) *$\mathcal{S}_\alpha(\boldsymbol{z}^\star) = \boldsymbol{0}$ if and only if $\boldsymbol{z}^\star$ is a KKT point of* (1)*.*

(ii) *If $\boldsymbol{z}^\star$ is both an isolated local minimizer and an isolated KKT point of* (1)*, then it is asymptotically stable relative to $\mathcal{C}$.*

The two claims follow from [10, Proposition 5.1 and the proof of Theorem 5.7(i)], respectively.

III. Distributed Algorithm

The proposed algorithm consists of an auxiliary projected saddle-point subsystem and output dynamics defined by local SGF-QPs.

For each coupling inequality $k = 1, \dots, p$, introduce the mismatch variables $\rho_i^k \in \mathbb{R}$, $i = 1, \dots, N$, and stack them across the agents as $\boldsymbol{\rho}^k := \mathrm{col}(\rho_1^k, \dots, \rho_N^k) \in \mathbb{R}^N$. Let $\boldsymbol{\chi}_i \in \mathbb{R}^n$ be an auxiliary primal variable and define $\boldsymbol{\chi} := \mathrm{col}(\boldsymbol{\chi}_1, \dots, \boldsymbol{\chi}_N) \in \mathbb{R}^{Nn}$. The dual variables are $\lambda_i^k \in \mathbb{R}_{\ge 0}$, $k = 1, \dots, p$, and $\boldsymbol{\nu}_i \in \mathbb{R}^n$, with $\boldsymbol{\lambda}^k := \mathrm{col}(\lambda_1^k, \dots, \lambda_N^k) \in \mathbb{R}^N_{\ge 0}$ and $\boldsymbol{\nu} := \mathrm{col}(\boldsymbol{\nu}_1, \dots, \boldsymbol{\nu}_N) \in \mathbb{R}^{Nn}$. For $\varepsilon > 0$ and $\tau > 0$, each agent $i = 1, \dots, N$ implements

$$\begin{aligned} \tau\dot{\boldsymbol{\chi}}_i =& -\nabla f_i(\boldsymbol{\chi}_i) - \sum_{k=1}^p \lambda_i^k \nabla g_i^k(\boldsymbol{\chi}_i) \\ & - \sum_{j \in \mathcal{N}_i} a_{ij}(\boldsymbol{\chi}_i - \boldsymbol{\chi}_j) - \sum_{j \in \mathcal{N}_i} a_{ij}(\boldsymbol{\nu}_i - \boldsymbol{\nu}_j), \\ \tau\dot{\rho}_i^k =& -\varepsilon\rho_i^k - \sum_{j \in \mathcal{N}_i} a_{ij}(\lambda_i^k - \lambda_j^k), \\ \tau\dot{\lambda}_i^k =& \Big[g_i^k(\boldsymbol{\chi}_i) + \sum_{j \in \mathcal{N}_i} a_{ij}(\rho_i^k - \rho_j^k)\Big]_{\lambda_i^k}^+, \\ \tau\dot{\boldsymbol{\nu}}_i =& \sum_{j \in \mathcal{N}_i} a_{ij}(\boldsymbol{\chi}_i - \boldsymbol{\chi}_j), \end{aligned} \tag{3}$$

where the second and third equations hold for $k = 1, \dots, p$. The auxiliary variables enter the output dynamics through the nominal direction

$$\boldsymbol{d}_i := \nabla f_i(\boldsymbol{x}_i) + \sum_{j \in \mathcal{N}_i} a_{ij}(\boldsymbol{x}_i - \boldsymbol{x}_j) + \sum_{j \in \mathcal{N}_i} a_{ij}(\boldsymbol{\nu}_i - \boldsymbol{\nu}_j). \tag{4}$$

To collect the mismatch variables stored by agent i, let $\boldsymbol{\rho}_i := \mathrm{col}(\rho_i^1, \dots, \rho_i^p) \in \mathbb{R}^p$ and

$$\boldsymbol{\zeta}_i := (\boldsymbol{x}_i, \boldsymbol{x}_{\mathcal{N}_i}, \boldsymbol{\rho}_i, \boldsymbol{\rho}_{\mathcal{N}_i}, \boldsymbol{\nu}_i, \boldsymbol{\nu}_{\mathcal{N}_i}).$$

For the current auxiliary and neighbor states, the local SGF-QP of agent i defines $S_i^\alpha(\boldsymbol{\zeta}_i)$ as the unique optimizer of

$$\begin{aligned} \min_{\boldsymbol{\xi}_i \in \mathbb{R}^n} \quad & \frac{1}{2}\|\boldsymbol{\xi}_i + \boldsymbol{d}_i\|^2 \\ \text{s.t.} \quad & \nabla g_i^k(\boldsymbol{x}_i)^\top \boldsymbol{\xi}_i \le -\alpha \left[g_i^k(\boldsymbol{x}_i) + \sum_{j \in \mathcal{N}_i} a_{ij}(\rho_i^k - \rho_j^k)\right], \\ & k = 1, \dots, p. \end{aligned} \tag{5}$$

The output trajectory is generated by

$$\dot{\boldsymbol{x}}_i = S_i^\alpha(\boldsymbol{\zeta}_i).$$

Remark 1. *Unlike [12], the local SGF-QP in the present method enforces only the localized coupling inequalities, while consensus is driven by the Laplacian terms in the nominal direction* (4)*. Consequently, consensus is required only asymptotically, and the output initialization need not satisfy consensus.*

This treatment of consensus creates an additional analytical issue. Both the auxiliary subsystem and the output dynamics depend on $\boldsymbol{\nu}$ only through $\mathcal{L}\boldsymbol{\nu}$, so components of $\boldsymbol{\nu}$ in $\ker(\mathcal{L})$ do not affect either subsystem. Since $\dot{\boldsymbol{\nu}} \in \mathrm{Range}(\mathcal{L})$, these components remain constant, and the equilibria of (3) are nonunique along $\ker(\mathcal{L})$. Consequently, the auxiliary subsystem has no unique equilibrium in the original coordinates, so the cascade argument in [12, Theorem 5.3] cannot be applied directly.

For analysis, define $\boldsymbol{w} := \mathcal{L}\boldsymbol{\nu} \in \mathrm{Range}(\mathcal{L})$ and $\boldsymbol{\eta} := \mathrm{col}(\boldsymbol{\chi}, \boldsymbol{\rho}, \boldsymbol{\lambda}, \boldsymbol{w})$, where $\boldsymbol{\rho} := \mathrm{col}(\boldsymbol{\rho}^1, \dots, \boldsymbol{\rho}^p)$ and $\boldsymbol{\lambda} := \mathrm{col}(\boldsymbol{\lambda}^1, \dots, \boldsymbol{\lambda}^p)$. Because $\tau\dot{\boldsymbol{w}} = \mathcal{L}^2\boldsymbol{\chi} \in \mathrm{Range}(\mathcal{L})$, $\boldsymbol{w}(t)$ remains in $\mathrm{Range}(\mathcal{L})$ and these variables form a closed subsystem. The algorithm admits the compact representation

$$\tau\dot{\boldsymbol{\eta}} = \Phi_\varepsilon(\boldsymbol{\eta}), \quad \dot{\boldsymbol{x}} = S_\alpha(\boldsymbol{x}, \boldsymbol{\eta}), \tag{6}$$

where Φ_ε denotes the closed vector field obtained from (3) together with $\tau\dot{\boldsymbol{w}} = \mathcal{L}^2\boldsymbol{\chi}$, and $S_\alpha := \mathrm{col}(S_1^\alpha, \dots, S_N^\alpha)$. The variable $\boldsymbol{w}$ is used only in the analysis. The distributed implementation remains (3).

IV. Convergence Analysis

This section establishes convergence of the auxiliary subsystem, analyzes the corresponding reduced-order model, and studies the cascaded system (6).

A. Auxiliary subsystem

Let $\boldsymbol{g}^k(\boldsymbol{\chi}) := \mathrm{col}(g_1^k(\boldsymbol{\chi}_1), \dots, g_N^k(\boldsymbol{\chi}_N))$. The auxiliary dynamics (3) is associated with

$$\begin{aligned} \min_{\boldsymbol{\chi}, \boldsymbol{\rho}} \quad & F(\boldsymbol{\chi}) + \frac{\varepsilon}{2}\sum_{k=1}^{p}\|\boldsymbol{\rho}^k\|^2 \\ \text{s.t.} \quad & \boldsymbol{g}^k(\boldsymbol{\chi}) + L\boldsymbol{\rho}^k \le \mathbf{0}, \quad k = 1, \dots, p, \\ & \mathcal{L}\boldsymbol{\chi} = \mathbf{0}. \end{aligned} \tag{P_ε}$$

The term $-\mathcal{L}\boldsymbol{\chi}$ in (3) is generated by the augmentation $\boldsymbol{\chi}^\top\mathcal{L}\boldsymbol{\chi}/2$. Since both the augmentation and its gradient vanish when $\mathcal{L}\boldsymbol{\chi} = \mathbf{0}$, adding this augmentation does not change the KKT points of (P_ε).

The regularized problem (P_ε) has a unique primal optimizer, but the rank-deficient Laplacian equality can still make its multiplier nonunique. The following assumption is used to establish uniqueness of $\boldsymbol{\lambda}^{\star,\varepsilon}$ and $\mathcal{L}\boldsymbol{\nu}^{\star,\varepsilon}$, where the latter enters the output dynamics through (4).

Assumption 4 (Aggregate LICQ)**.** *Fix $\varepsilon > 0$. Let $(\boldsymbol{\chi}^{\star,\varepsilon}, \boldsymbol{\rho}^{\star,\varepsilon})$ be the primal optimizer of (P_ε), and write $\boldsymbol{\chi}^{\star,\varepsilon} = \mathbf{1}_N \otimes \boldsymbol{z}^{\star,\varepsilon}$. Define $\mathcal{I}_\varepsilon^\star := \{k : G_k(\boldsymbol{\chi}^{\star,\varepsilon}) = 0\}$. LICQ holds for the active aggregate inequality constraints at $\boldsymbol{z}^{\star,\varepsilon}$, namely, the vectors*

$$\left\{\sum_{i=1}^{N}\nabla g_i^k(\boldsymbol{z}^{\star,\varepsilon}) : k \in \mathcal{I}_\varepsilon^\star\right\}$$

are linearly independent.

Proposition 2. *Under Assumptions 1–4, the following hold:*

(i) *Every equilibrium of* (3) *induces the same equilibrium of the $\boldsymbol{\eta}$-dynamics, given by*

$$\boldsymbol{\eta}^{\star,\varepsilon} := \mathrm{col}(\boldsymbol{\chi}^{\star,\varepsilon}, \boldsymbol{\rho}^{\star,\varepsilon}, \boldsymbol{\lambda}^{\star,\varepsilon}, \boldsymbol{w}^{\star,\varepsilon}), \quad \boldsymbol{w}^{\star,\varepsilon} := \mathcal{L}\boldsymbol{\nu}^{\star,\varepsilon}.$$

(ii) *For every initial condition of* (3) *satisfying $\boldsymbol{\lambda}(0) \ge \mathbf{0}$, there exists a unique Carathéodory solution on $[0, \infty)$, and the induced trajectory satisfies*

$$\boldsymbol{\eta}(t) \longrightarrow \boldsymbol{\eta}^{\star,\varepsilon}. \tag{7}$$

(iii) *The equilibrium $\boldsymbol{\eta}^{\star,\varepsilon}$ is globally asymptotically stable for the $\boldsymbol{\eta}$-dynamics restricted to $\boldsymbol{\lambda} \ge \mathbf{0}$ and $\boldsymbol{w} \in \mathrm{Range}(\mathcal{L})$.*

Proof. Define $\widetilde{\boldsymbol{\chi}} := \mathbf{1}_N \otimes \bar{\boldsymbol{z}}$, where $\bar{\boldsymbol{z}}$ is given by Assumption 3. For each $k = 1, \dots, p$, $\mathrm{Range}(L) = \mathbf{1}_N^\perp$ allows one to choose $\widetilde{\boldsymbol{\rho}}^k$ so that every component of $\boldsymbol{g}^k(\widetilde{\boldsymbol{\chi}}) + L\widetilde{\boldsymbol{\rho}}^k$ equals $G_k(\widetilde{\boldsymbol{\chi}})/N < 0$. Hence ($P_\varepsilon$) satisfies Slater's condition. Its feasible set is nonempty and closed, and its objective is strongly convex in $(\boldsymbol{\chi}, \boldsymbol{\rho})$. Therefore, ($P_\varepsilon$) admits a unique primal optimizer. Let $q_i^{k,\star} := g_i^k(\boldsymbol{\chi}_i^{\star,\varepsilon}) + (L\boldsymbol{\rho}^{k,\star,\varepsilon})_i$. For all i and k, its KKT conditions are

$$\begin{aligned} &\mathbf{0} = \nabla f_i(\boldsymbol{\chi}_i^{\star,\varepsilon}) + \sum_{k=1}^{p}\lambda_i^{k,\star,\varepsilon}\nabla g_i^k(\boldsymbol{\chi}_i^{\star,\varepsilon}) + (\boldsymbol{w}^{\star,\varepsilon})_i, \\ &\mathbf{0} = \varepsilon\boldsymbol{\rho}^{k,\star,\varepsilon} + L\boldsymbol{\lambda}^{k,\star,\varepsilon}, \\ &q_i^{k,\star} \le 0, \quad \lambda_i^{k,\star,\varepsilon} \ge 0, \quad \lambda_i^{k,\star,\varepsilon}q_i^{k,\star} = 0, \quad \mathcal{L}\boldsymbol{\chi}^{\star,\varepsilon} = \mathbf{0}. \end{aligned}$$

Consider two pairs of KKT multipliers $(\boldsymbol{\lambda}^{(1)}, \boldsymbol{\nu}^{(1)})$ and $(\boldsymbol{\lambda}^{(2)}, \boldsymbol{\nu}^{(2)})$ associated with the same primal optimizer. Define $\Delta\boldsymbol{\lambda}^k := \boldsymbol{\lambda}^{k,(1)} - \boldsymbol{\lambda}^{k,(2)}$ and $\Delta\boldsymbol{\nu} := \boldsymbol{\nu}^{(1)} - \boldsymbol{\nu}^{(2)}$. The $\boldsymbol{\rho}^k$-stationarity equations yield $L\Delta\boldsymbol{\lambda}^k = \mathbf{0}$, and hence $\Delta\boldsymbol{\lambda}^k = a_k\mathbf{1}_N$ for some scalar a_k. If any localized constraint associated with index k is inactive, complementarity implies that the corresponding multiplier is zero for both KKT pairs, and therefore $a_k = 0$. Hence, only indices in $\mathcal{I}_\varepsilon^\star$ can have $a_k \ne 0$.

For each $i = 1, \dots, N$, subtracting the $\boldsymbol{\chi}$-stationarity equations gives

$$\sum_{k \in \mathcal{I}_\varepsilon^\star} a_k\nabla g_i^k(\boldsymbol{z}^{\star,\varepsilon}) + (\mathcal{L}\Delta\boldsymbol{\nu})_i = \mathbf{0}.$$

Summing over i yields

$$\sum_{k \in \mathcal{I}_\varepsilon^\star} a_k\sum_{i=1}^{N}\nabla g_i^k(\boldsymbol{z}^{\star,\varepsilon}) = \mathbf{0}.$$

Assumption 4 then gives $a_k = 0$ for every $k \in \mathcal{I}_\varepsilon^\star$. Together with the inactive-constraint case, this proves that $\boldsymbol{\lambda}^{\star,\varepsilon}$ is unique. The $\boldsymbol{\chi}$-stationarity equations subsequently imply that $\mathcal{L}\boldsymbol{\nu}^{\star,\varepsilon}$ is also unique.

For convergence, [13, Lemmas 4.1–4.4] ensures that the projected dynamics preserves $\boldsymbol{\lambda} \ge \mathbf{0}$ and has a unique local Carathéodory solution. The remaining dynamics is locally Lipschitz. Moreover, $\dot{\boldsymbol{\nu}} \in \mathrm{Range}(\mathcal{L})$, so its $\ker(\mathcal{L})$ component is constant. Since $\mathcal{L}\boldsymbol{\nu}^{\star,\varepsilon}$ is unique, we may select a KKT

multiplier $\boldsymbol{\nu}^{\star,\varepsilon}$ with the same kernel component as $\boldsymbol{\nu}(0)$ and consider

$$V := \frac{1}{2}\big(\|\boldsymbol{\chi}-\boldsymbol{\chi}^{\star,\varepsilon}\|^2 + \|\boldsymbol{\rho}-\boldsymbol{\rho}^{\star,\varepsilon}\|^2 + \|\boldsymbol{\lambda}-\boldsymbol{\lambda}^{\star,\varepsilon}\|^2 + \|\boldsymbol{\nu}-\boldsymbol{\nu}^{\star,\varepsilon}\|^2\big). \tag{8}$$

The projection inequality and the convexity argument in [13, Theorem 4.5] give

$$\tau\dot{V} \le -\sigma\|\boldsymbol{\chi}-\boldsymbol{\chi}^{\star,\varepsilon}\|^2 - \varepsilon\|\boldsymbol{\rho}-\boldsymbol{\rho}^{\star,\varepsilon}\|^2 - \boldsymbol{\chi}^\top\mathcal{L}\boldsymbol{\chi} \le 0. \tag{9}$$

The equality-related cross terms cancel because $\mathcal{L}$ is symmetric. Hence solutions remain in compact sublevel sets of V and are complete. On $\{\dot{V}=0\}$, strong convexity gives $\boldsymbol{\chi}=\boldsymbol{\chi}^{\star,\varepsilon}$ and $\boldsymbol{\rho}=\boldsymbol{\rho}^{\star,\varepsilon}$. Invariance then forces the remaining KKT conditions. The multiplier uniqueness proved above and the fixed kernel component of $\boldsymbol{\nu}$ reduce the largest invariant subset to the selected KKT point. LaSalle's argument in [13, Theorem 4.5] therefore gives (7). Finally, $\mathcal{L}$ is one-to-one on $\mathrm{Range}(\mathcal{L})$, so the norms of the range components of $\boldsymbol{\nu}-\boldsymbol{\nu}^{\star,\varepsilon}$ and $\mathcal{L}(\boldsymbol{\nu}-\boldsymbol{\nu}^{\star,\varepsilon})$ are equivalent. Together with (9), this proves stability and hence global asymptotic stability of the reduced $\boldsymbol{\eta}$-equilibrium. □

B. Anytime feasibility and regularization error

Let $\mathcal{D}$ be the set of $(\boldsymbol{x},\boldsymbol{\eta})$ for which every local SGF-QP in (5) satisfies Slater's condition and LICQ at its optimizer. The strongly convex QP objective and the parametric-QP arguments in [10, Proposition 5.3] and [12, Proposition 4.6] show that $\mathcal{D}$ is open, and that S_α and its unique multiplier map are locally Lipschitz there.

Lemma 1. *Under Assumptions 1 and 2, the following hold for each $k=1,\dots,p$.*

(i) *$G_k(\boldsymbol{x}) \le 0$ if and only if there exists $\widehat{\boldsymbol{\rho}}^k \in \mathbb{R}^N$ such that $g_i^k(\boldsymbol{x}_i) + (L\widehat{\boldsymbol{\rho}}^k)_i \le 0$ for all $i=1,\dots,N$.*

(ii) *Along any solution of* (6) *contained in $\mathcal{D}$,*

$$G_k(\boldsymbol{x}(t)) \le e^{-\alpha t}G_k(\boldsymbol{x}(0)). \tag{10}$$

Proof. For (i), summing the localized inequalities proves one direction. Conversely, set $\bar{g}^k := G_k(\boldsymbol{x})/N$ and $\boldsymbol{r}^k := \mathrm{col}(g_1^k(\boldsymbol{x}_1),\dots,g_N^k(\boldsymbol{x}_N)) - \bar{g}^k\mathbf{1}_N$. Since $\mathbf{1}_N^\top\boldsymbol{r}^k = 0$, there exists $\widehat{\boldsymbol{\rho}}^k$ satisfying $L\widehat{\boldsymbol{\rho}}^k = -\boldsymbol{r}^k$, which gives every localized constraint the value $\bar{g}^k$.

For (ii), the local SGF-QP constraints yield

$$\dot{G}_k(\boldsymbol{x}) \le -\alpha\sum_{i=1}^N \big(g_i^k(\boldsymbol{x}_i) + (L\boldsymbol{\rho}^k)_i\big) = -\alpha G_k(\boldsymbol{x}).$$

The comparison lemma [14, Lemma 3.4] gives (10). □

Lemma 2. *Under Assumptions 1–3, let $\boldsymbol{x}^\star$ be the optimizer of* (P). *Fix any $\bar{\boldsymbol{\rho}}$, independent of ε, such that $(\boldsymbol{x}^\star,\bar{\boldsymbol{\rho}})$ is feasible for (P_ε). Then*

$$\|\boldsymbol{\chi}^{\star,\varepsilon}-\boldsymbol{x}^\star\| \le \left(\frac{\varepsilon}{\sigma}\sum_{k=1}^p\|\bar{\boldsymbol{\rho}}^k\|^2\right)^{1/2}.$$

Consequently, $\boldsymbol{\chi}^{\star,\varepsilon}\to\boldsymbol{x}^\star$ as $\varepsilon\to 0$.

Proof. Lemma 1 identifies the projection of the feasible set of (P_ε) onto $\boldsymbol{\chi}$ with the feasible set of (P). Since this set is convex and $\boldsymbol{x}^\star$ minimizes F over it, the first-order optimality condition gives $\nabla F(\boldsymbol{x}^\star)^\top(\boldsymbol{\chi}^{\star,\varepsilon}-\boldsymbol{x}^\star)\ge 0$. Strong convexity and optimality in (P_ε) then give

$$\frac{\sigma}{2}\|\boldsymbol{\chi}^{\star,\varepsilon}-\boldsymbol{x}^\star\|^2 \le F(\boldsymbol{\chi}^{\star,\varepsilon}) - F(\boldsymbol{x}^\star) \le \frac{\varepsilon}{2}\sum_{k=1}^p\|\bar{\boldsymbol{\rho}}^k\|^2.$$

This proves the result. □

C. Reduced-order model and cascade convergence

Setting $\boldsymbol{\eta}=\boldsymbol{\eta}^{\star,\varepsilon}$ freezes the auxiliary subsystem at its equilibrium and removes its state from the cascaded dynamics. The resulting system, referred to as the reduced-order model, is the SGF associated with

$$\begin{aligned}\min_{\boldsymbol{x}}\quad & F(\boldsymbol{x}) + (\boldsymbol{w}^{\star,\varepsilon})^\top\boldsymbol{x} + \frac{1}{2}\boldsymbol{x}^\top\mathcal{L}\boldsymbol{x}\\ \text{s.t.}\quad & g_i^k(\boldsymbol{x}_i) + (L\boldsymbol{\rho}^{k,\star,\varepsilon})_i \le 0.\end{aligned} \tag{R_ε}$$

Let ψ_ε denote the objective of (R_ε). The inequalities in (R_ε) are indexed by $i=1,\dots,N$ and $k=1,\dots,p$. Its feasible set is

$$\mathcal{X}_\varepsilon^{\mathrm{red}} := \{\boldsymbol{x} : g_i^k(\boldsymbol{x}_i) + (L\boldsymbol{\rho}^{k,\star,\varepsilon})_i \le 0,\ i=1,\dots,N,\quad k=1,\dots,p\}.$$

Lemma 1 gives $\mathcal{X}_\varepsilon^{\mathrm{red}} \subset \mathcal{X}_G$. Let $\bar{\boldsymbol{x}}$ denote the state of the reduced-order model:

$$\dot{\bar{\boldsymbol{x}}} = S_\alpha(\bar{\boldsymbol{x}},\boldsymbol{\eta}^{\star,\varepsilon}). \tag{11}$$

Lemma 3. *Under Assumptions 1–4, the optimizer $\boldsymbol{\chi}^{\star,\varepsilon}$ of (P_ε) is also the unique optimizer of (R_ε). If $\mathcal{X}_\varepsilon^{\mathrm{red}}\times\{\boldsymbol{\eta}^{\star,\varepsilon}\}\subset\mathcal{D}$, then $\boldsymbol{\chi}^{\star,\varepsilon}$ is the unique equilibrium of* (11) *and is globally asymptotically stable relative to $\mathcal{X}_\varepsilon^{\mathrm{red}}$.*

Proof. At the KKT point of (P_ε), fix $\boldsymbol{\rho}=\boldsymbol{\rho}^{\star,\varepsilon}$ and $\boldsymbol{w}^{\star,\varepsilon}=\mathcal{L}\boldsymbol{\nu}^{\star,\varepsilon}$. Because $\mathcal{L}\boldsymbol{\chi}^{\star,\varepsilon}=\mathbf{0}$, the KKT conditions of (P_ε) reduce to those of (R_ε). Thus $\boldsymbol{\chi}^{\star,\varepsilon}$ is its unique optimizer by strong convexity.

The inclusion in $\mathcal{D}$ gives local Lipschitzness and a strictly feasible local direction for every active constraint. Stacking these directions verifies MFCQ for (R_ε). Proposition 1(i) then shows that $\boldsymbol{\chi}^{\star,\varepsilon}$ is the unique equilibrium, while Proposition 1(ii) gives its asymptotic stability relative to $\mathcal{X}_\varepsilon^{\mathrm{red}}$. The SGF constraints also make this set forward invariant, and [10, Lemma 5.8] gives

$$\dot{\psi}_\varepsilon(\bar{\boldsymbol{x}}) \le -\|S_\alpha(\bar{\boldsymbol{x}},\boldsymbol{\eta}^{\star,\varepsilon})\|^2.$$

Strong convexity makes the sublevel sets compact. LaSalle's invariance principle [15] therefore gives convergence of every feasible trajectory to the unique equilibrium, proving the claim. □

Let $\mathcal{K}_{\eta,0}$ be a compact set of initial conditions for the $\boldsymbol{\eta}$-dynamics with $\boldsymbol{\lambda}(0)\ge\mathbf{0}$ and $\boldsymbol{w}(0)\in\mathrm{Range}(\mathcal{L})$. Equations (8) and (9), together with the compactness of $\mathcal{K}_{\eta,0}$, show that all $\boldsymbol{\eta}$-trajectories initialized in $\mathcal{K}_{\eta,0}$ are contained

in a common compact set, denoted by $\mathcal{K}_\eta$. For a compact $\Omega_x \subset \mathcal{X}_\varepsilon^{\mathrm{red}}$, define

$$\mathcal{K}_\Omega := \left\{ \boldsymbol{x} \in \mathcal{X}_\varepsilon^{\mathrm{red}} : \psi_\varepsilon(\boldsymbol{x}) \le \max_{\boldsymbol{z} \in \Omega_x} \psi_\varepsilon(\boldsymbol{z}) \right\}.$$

This compact sublevel set contains every trajectory of (11) initialized in Ω_x.

Proposition 3. *Let $\boldsymbol{\eta}(0) \in \mathcal{K}_{\eta,0}$. Under Assumptions 1–4, the following hold:*

(i) *For every compact $\Omega_x \subset \mathcal{X}_\varepsilon^{\mathrm{red}}$ satisfying $\mathcal{K}_\Omega \times \mathcal{K}_\eta \subset \mathcal{D}$ and every $\delta > 0$, there exists $\tau_{\delta,\Omega} > 0$ such that $0 < \tau < \tau_{\delta,\Omega}$ implies*

$$\limsup_{t \to \infty} \|\boldsymbol{x}(t) - \boldsymbol{\chi}^{\star,\varepsilon}\| \le \delta \tag{12}$$

whenever $\boldsymbol{x}(0) \in \Omega_x$.

(ii) *If $\mathcal{X}_G$ is bounded, $\mathcal{X}_G \times \mathcal{K}_\eta \subset \mathcal{D}$, and $\boldsymbol{x}(0) \in \mathcal{X}_G$, then, for every $\tau > 0$,*

$$\boldsymbol{x}(t) \longrightarrow \boldsymbol{\chi}^{\star,\varepsilon}. \tag{13}$$

Proof. For (i), we apply the singular-perturbation argument in [12, Theorem 5.3]. Proposition 2 and Lemma 3 provide the globally asymptotically stable equilibria of the auxiliary and reduced systems on the stated invariant sets. Moreover, ψ_ε is nonincreasing along (11), and the compact inclusion $\mathcal{K}_\Omega \times \mathcal{K}_\eta \subset \mathcal{D}$ gives the required local Lipschitzness. Since $\mathcal{D}$ is open and $\boldsymbol{\chi}^{\star,\varepsilon} \in \mathcal{K}_\Omega$, one may slightly enlarge $\mathcal{K}_\Omega$ inside $\mathcal{D}$ so that it contains the sufficiently small target neighborhood required in that theorem. Applying it with a radius no larger than δ yields (12).

For (ii), we follow the second part of the proof in [12, Theorem 5.3]. Lemma 1 keeps $\boldsymbol{x}(t)$ in the compact set $\mathcal{X}_G$, while Proposition 2 gives $\boldsymbol{\eta}(t) \to \boldsymbol{\eta}^{\star,\varepsilon}$. Using the $\boldsymbol{\rho}$-dynamics in (3) and the equilibrium condition $\varepsilon \boldsymbol{\rho}^{k,\star,\varepsilon} + L\boldsymbol{\lambda}^{k,\star,\varepsilon} = \mathbf{0}$, we also obtain $\dot{\boldsymbol{\rho}}(t) \to \mathbf{0}$.

For $i = 1, \dots, N$ and $k = 1, \dots, p$, define

$$c_i^k(t) := g_i^k(\boldsymbol{x}_i(t)) + (L\boldsymbol{\rho}^k(t))_i,$$
$$c_{i,\star}^k(\boldsymbol{x}) := g_i^k(\boldsymbol{x}_i) + (L\boldsymbol{\rho}^{k,\star,\varepsilon})_i.$$

The local SGF-QP constraints give

$$\dot{c}_i^k(t) \le -\alpha c_i^k(t) + (L\dot{\boldsymbol{\rho}}^k(t))_i.$$

The comparison lemma [14, Lemma 3.4] and $\dot{\boldsymbol{\rho}}(t) \to \mathbf{0}$ imply

$$\limsup_{t \to \infty} c_i^k(t) \le 0. \tag{14}$$

Since $\boldsymbol{\rho}(t) \to \boldsymbol{\rho}^{\star,\varepsilon}$, it follows that

$$\limsup_{t \to \infty} c_{i,\star}^k(\boldsymbol{x}(t)) \le 0. \tag{15}$$

Since $\boldsymbol{x}(t)$ remains in the compact set $\mathcal{X}_G$, (15) implies that every accumulation point of $\boldsymbol{x}(t)$ belongs to $\mathcal{X}_\varepsilon^{\mathrm{red}}$. Moreover, since $\boldsymbol{\eta}(t) \to \boldsymbol{\eta}^{\star,\varepsilon}$, Lemma 3 and compactness of $\mathcal{X}_G \times \mathcal{K}_\eta$ imply that, outside any fixed neighborhood of $\boldsymbol{\chi}^{\star,\varepsilon}$, $\|S_\alpha(\boldsymbol{x}(t), \boldsymbol{\eta}(t))\|$ has a positive lower bound for all sufficiently large t.

For each local SGF-QP at time t, let $\varphi_i^k(t) \ge 0$ be an associated KKT multiplier. Let $\boldsymbol{d} := \mathrm{col}(\boldsymbol{d}_1, \dots, \boldsymbol{d}_N)$, where $\boldsymbol{d}_i$ is defined in (4). Along the $\boldsymbol{x}$-dynamics in (6),

$$\begin{aligned}
\dot{\psi}_\varepsilon(\boldsymbol{x}) &= \big(\boldsymbol{d} + \boldsymbol{w}^{\star,\varepsilon} - \boldsymbol{w}\big)^\top S_\alpha(\boldsymbol{x}, \boldsymbol{\eta}) \\
&= -\|S_\alpha(\boldsymbol{x}, \boldsymbol{\eta})\|^2 - \sum_{i=1}^N \sum_{k=1}^p \varphi_i^k(t) \nabla g_i^k(\boldsymbol{x}_i)^\top S_i^\alpha(\boldsymbol{\zeta}_i) \\
&\quad + (\boldsymbol{w}^{\star,\varepsilon} - \boldsymbol{w})^\top S_\alpha(\boldsymbol{x}, \boldsymbol{\eta}) \\
&= -\|S_\alpha(\boldsymbol{x}, \boldsymbol{\eta})\|^2 + \alpha \sum_{i=1}^N \sum_{k=1}^p \varphi_i^k(t) c_i^k(t) \\
&\quad + (\boldsymbol{w}^{\star,\varepsilon} - \boldsymbol{w})^\top S_\alpha(\boldsymbol{x}, \boldsymbol{\eta}).
\end{aligned} \tag{16}$$

The last two equalities use the stationarity condition of each local SGF-QP, $S_i^\alpha(\boldsymbol{\zeta}_i) + \boldsymbol{d}_i + \sum_{k=1}^p \varphi_i^k(t) \nabla g_i^k(\boldsymbol{x}_i) = \mathbf{0}$, and the complementarity condition $\varphi_i^k(t)\big(\nabla g_i^k(\boldsymbol{x}_i)^\top S_i^\alpha(\boldsymbol{\zeta}_i) + \alpha c_i^k(t)\big) = 0$ for all i and k.

The first term $-\|S_\alpha(\boldsymbol{x}, \boldsymbol{\eta})\|^2$ in (16) is nonpositive. Since the multiplier map is locally Lipschitz on $\mathcal{D}$, it is uniformly bounded on the compact set $\mathcal{X}_G \times \mathcal{K}_\eta$. Together with (14), this shows that the middle term in (16) has limit superior at most zero. Moreover, the last term converges to zero because $\boldsymbol{w}(t) \to \boldsymbol{w}^{\star,\varepsilon}$ and S_α is bounded on $\mathcal{X}_G \times \mathcal{K}_\eta$. Therefore,

$$\begin{aligned}
\limsup_{t \to \infty} \Bigg[&\alpha \sum_{i=1}^N \sum_{k=1}^p \varphi_i^k(t) c_i^k(t) \\
&+ (\boldsymbol{w}^{\star,\varepsilon} - \boldsymbol{w}(t))^\top S_\alpha(\boldsymbol{x}(t), \boldsymbol{\eta}(t)) \Bigg] \le 0.
\end{aligned} \tag{17}$$

Combining the fact that $\|S_\alpha(\boldsymbol{x}(t), \boldsymbol{\eta}(t))\|$ has a positive lower bound outside any fixed neighborhood of $\boldsymbol{\chi}^{\star,\varepsilon}$ for all sufficiently large t with (17) in (16), we obtain $\dot{\psi}_\varepsilon(\boldsymbol{x}(t)) < 0$ outside that neighborhood for all sufficiently large t. Following the second part of the proof of [12, Theorem 5.3], the trajectory eventually enters and remains in the neighborhood. Since the neighborhood is arbitrary, (13) follows for every fixed $\tau > 0$. □

Theorem 1. *Fix $\varepsilon > 0$, let $\boldsymbol{\eta}(0) \in \mathcal{K}_{\eta,0}$, and suppose Assumptions 1–4 hold. Then the following hold:*

(i) ***Anytime coupling inequalities.*** *Along every solution of (6) contained in $\mathcal{D}$, $\boldsymbol{x}(0) \in \mathcal{X}_G$ implies $\boldsymbol{x}(t) \in \mathcal{X}_G$ for all $t \ge 0$.*

(ii) ***Asymptotic consensus.*** *If $\mathcal{X}_G$ is bounded, $\mathcal{X}_G \times \mathcal{K}_\eta \subset \mathcal{D}$, and $\boldsymbol{x}(0) \in \mathcal{X}_G$, then, for every $\tau > 0$,*

$$\boldsymbol{x}(t) \to \boldsymbol{\chi}^{\star,\varepsilon}, \quad \mathcal{L}\boldsymbol{x}(t) \to \mathbf{0}.$$

Proof. Part (i) is Lemma 1(ii). Part (ii) follows from Proposition 3(ii) and $\mathcal{L}\boldsymbol{\chi}^{\star,\varepsilon} = \mathbf{0}$. □

Lemma 2 and Proposition 3 give the asymptotic error bound $\delta + (\varepsilon \sum_k \|\bar{\boldsymbol{\rho}}^k\|^2/\sigma)^{1/2}$ in case (i), and the same bound with $\delta = 0$ in case (ii). Thus the regularization error vanishes as $\varepsilon \to 0$.

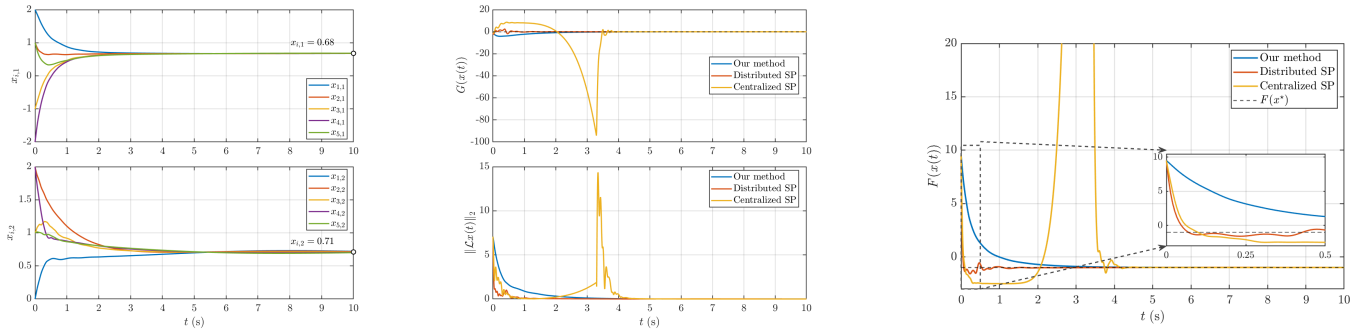


Fig. 1. Simulation results. Left: agent states. Middle: coupling constraint and consensus error. Right: objective values with the initial transient enlarged.

V. Numerical Experiments

We consider five agents with $\boldsymbol{x}_i = \mathrm{col}(x_{i,1}, x_{i,2}) \in \mathbb{R}^2$ over the unit-weight path graph 1–2–3–4–5. Its Laplacian matrix is

$$L = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}, \quad \mathcal{L} = L \otimes I_2.$$

We set $f_i(\boldsymbol{x}_i) = \frac{1}{2}\|\boldsymbol{x}_i\|^2 - x_{i,1}$ and $g_i(\boldsymbol{x}_i) = 1 + p_{i,1}x_{i,1}^2 - p_{i,2}x_{i,2}$, so that $G(\boldsymbol{x}) := \sum_{i=1}^5 g_i(\boldsymbol{x}_i) \le 0$. The parameters are $\boldsymbol{p}_1 = \mathrm{col}(0, 2)$, $\boldsymbol{p}_2 = \mathrm{col}(0.5, 3)$, $\boldsymbol{p}_3 = \mathrm{col}(0.5, -1)$, $\boldsymbol{p}_4 = \mathrm{col}(1, 2)$, and $\boldsymbol{p}_5 = \mathrm{col}(1, 3)$.

We initialize $\boldsymbol{x}_1(0) = \mathrm{col}(2, 0)$, $\boldsymbol{x}_2(0) = \mathrm{col}(1, 2)$, $\boldsymbol{x}_3(0) = \mathrm{col}(-1, 1)$, $\boldsymbol{x}_4(0) = \mathrm{col}(-2, 2)$, and $\boldsymbol{x}_5(0) = \mathrm{col}(1, 1)$. Thus $G(\boldsymbol{x}(0)) = -1 < 0$, but $\|\mathcal{L}\boldsymbol{x}(0)\| = \sqrt{50}$, so consensus is not initially satisfied. We further set $\boldsymbol{\chi}(0) = \boldsymbol{x}(0)$, $\boldsymbol{\rho}(0) = \boldsymbol{\lambda}(0) = \boldsymbol{\nu}(0) = \mathbf{0}$, $\alpha = 1$, $\varepsilon = 10^{-3}$, and $\tau = 0.1$.

We compare with two projected saddle-point baselines without SGF output: Distributed SP uses $\boldsymbol{\chi}(t)$ from (3), while Centralized SP is applied directly to (P), treating $\mathcal{L}\boldsymbol{x} = \mathbf{0}$ as an equality constraint.

Figure 1 shows that the proposed output approaches the regularized optimizer while maintaining $G(\boldsymbol{x}(t)) \le 0$. Both baselines violate it transiently, although all three methods asymptotically reach consensus.

The value $F(\boldsymbol{x}(t))$ may temporarily fall below $F(\boldsymbol{x}^\star)$ because the transient state need not satisfy consensus and is therefore not necessarily feasible for the original problem. The method in [12] is not included because its anytime guarantee requires the initial output to satisfy consensus, whereas ours does not.

VI. Conclusion

We developed a distributed method that preserves coupling inequalities throughout the trajectory and reaches consensus asymptotically. The output can approach the regularized optimizer arbitrarily closely and converges to it under additional boundedness conditions, while the regularization error is $O(\sqrt{\varepsilon})$. Future work may investigate extensions to time-varying directed networks and more general coupling constraints.

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